

2024, Mar. 2024 Release LOC:Acute Adult

Infection: General

Utilization Benchmarks: Multiple options available, see table below for options

Overview

Select Day

Initial review ^(1, 2)

Episode Day 1 ⁽¹⁾

Episode Day 2 ⁽⁵⁵⁾

Episode Day 3 ⁽⁷¹⁾

Episode Day 4-6

Episode Day 7

Discharge Screens ⁽⁷⁸⁾

Utilization Benchmarks

Notes

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Overview

Instruction:

This subset is for the review of infections that are not included in condition-specific infection subsets. This includes criteria for patients with infections that have failed outpatient management and for patients with the following specific infections:

- Central venous catheter infection
- Dengue
- Epiglottitis
- Fever of unknown origin
- Influenza
- Intraocular infection
- Laboratory test indicative of infection
- Mastoiditis
- Mononucleosis
- Neutropenia
- New onset fever in the hospitalized patient
- Peritonsillar abscess
- Pregnancy with a suspected infection
- Retropharyngeal abscess
- Tuberculosis
- Typhoid
- Viral respiratory infection (excludes COVID-19)

For patients who begin to exhibit signs and symptoms consistent with sepsis and need additional interventions which are not provided in the current subset, please refer to the Infection: Sepsis subset.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The

Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, ≥ One:** ⁽³⁾● Dengue actual or suspected and, ≥ **One:** ^(4, 5)– Inadequate oral intake ⁽⁶⁾

– Age > 65 yrs

● Comorbidity, ≥ **One:**

– Diabetes mellitus

– Chronic renal disease

– Hemolytic disease

– Obesity

– Emergency services remote

– Follow-up care unreliable within 24h ⁽⁷⁾● Mononucleosis and, **Both:** ⁽⁸⁾

– Impaired swallowing

– Inadequate oral intake ⁽⁶⁾– Peritonsillar abscess actual or suspected ⁽⁹⁾● Viral respiratory infection actual or suspected and, **All:**

– Dyspnea

● Oxygenation, ≥ **One:**– Arterial Po₂ 56-60 mmHg(7.4-8.0 kPa) and < baseline ⁽¹⁰⁾– O₂ sat 90-91%(0.90-0.91) and < baseline ⁽¹⁰⁾– Requiring supplemental oxygen ⁽¹¹⁾● **ACUTE, ≥ One:** ⁽¹²⁾● Central venous catheter and, ≥ **One:** ⁽¹³⁾– Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾

– Outpatient blood culture positive

– Purulent drainage at exit site ⁽¹⁵⁾● Dengue actual or suspected and, ≥ **One:** ^(4, 16)

– Abdominal pain

– Ascites

– Hct > 10%(0.10) increase from baseline

– Hepatomegaly > 2 cm

– Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 ^(17, 18)

– Mucosal bleeding

– Platelet count < 100,000/cu.mm(100x10⁹/L)– Petechiae ⁽¹⁹⁾

– Pleural effusion

– Pregnancy

– Vomiting, protracted ⁽²⁰⁾● Fever of unknown origin and, **Both:** ⁽²¹⁾

– Temperature > 101.0°F(38.3°C) ≥ 3 wks

– Outpatient work-up non-diagnostic ⁽²²⁾– Intraocular infection and anti-infective ⁽²³⁾● Laboratory result indicative of infection, **Both:**● Test, ≥ **One:**

– Positive culture for bacteria

– Positive smear for bacteria

– Positive antigen for bacteria

– Positive latex agglutination for bacteria

- Positive culture for fungi
- Positive smear for fungi
- Positive antigen for fungi
- Positive latex agglutination for fungi
- Positive culture for protozoa
- Positive smear for protozoa
- Positive antigen for protozoa
- Positive latex agglutination for protozoa
- **Source, ≥ One:**
 - Biopsy, surgical (excludes skin) ⁽²⁴⁾
 - Blood
 - Bone marrow
 - Effluent smear ⁽²⁵⁾
- **Mastoiditis and, All:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
 - Confirmed by x-ray or CT
 - Erythema or pain over mastoid area
- **Neutropenia, Both:** ⁽²⁶⁾
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
- **ANC, One:** ⁽²⁷⁾
 - < 500/cu.mm(500x10⁶/L)
 - < 1000/cu.mm(1000x10⁶/L) with documented decline ⁽²⁸⁾
- **Pregnancy and suspected infection and, Both:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
- **WBC, ≥ One:**
 - > 18,000/cu.mm(18x10⁹/L)
 - > 15,000/cu.mm(15x10⁹/L) and bands > 7%(0.07)
- Retropharyngeal abscess actual or suspected ⁽²⁹⁾
- **TB actual or suspected and, Both:**
 - Negative airflow isolation
- **Finding, One:**
 - Public health risk ⁽³⁰⁾
 - < 3 consecutive negative AFB sputum smears ⁽³¹⁾
 - Positive AFB sputum smear
- **Typhoid fever actual or suspected and, Both:** ⁽³²⁾
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
- **Finding, ≥ One:**
 - Abdominal distention
 - Abdominal pain
 - Inadequate oral intake ⁽⁶⁾
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 ^(17, 18)
- **Viral respiratory infection actual or suspected and, All:**
 - Dyspnea
- **Oxygenation, ≥ One:**
 - Arterial Po₂ < 56 mmHg(7.4 kPa) and < baseline ⁽¹⁰⁾
 - O₂ sat ≤ 89%(0.89) and < baseline ⁽¹⁰⁾
 - Requiring supplemental oxygen ⁽¹¹⁾
- **INTERMEDIATE, ≥ One:** ⁽³³⁾
 - NIPPV ⁽³⁴⁾
- **Oxygen, ≥ One:** ⁽¹¹⁾
 - HFNC and respiratory monitoring every 3-4h ^(35, 36)
 - Oxygen ≥ 40%(0.40) ≤ 2d
- **CRITICAL, ≥ One:** ⁽³⁷⁾

- Epiglottitis actual or suspected and, **Both:**
 - Risk of airway compromise
- Finding, **≥ One:**
 - Confirmed by x-ray
 - Difficulty swallowing
 - Drooling
- Peritonsillar abscess actual or suspected and, **Both:** ⁽⁹⁾
 - Risk of airway compromise
- Finding, **≥ One:**
 - Confirmed by x-ray or CT
 - Difficulty swallowing
 - Drooling
- Retropharyngeal abscess actual or suspected and, **Both:** ⁽²⁹⁾
 - Risk of airway compromise
- Finding, **≥ One:**
 - Confirmed by x-ray or CT
 - Difficulty swallowing
 - Drooling
- Typhoid fever actual or suspected and, **≥ One:** ⁽³²⁾
 - Gastrointestinal bleeding
- Mental status change, **One:** ⁽¹⁷⁾
 - Finding, **≥ One:**
 - Coma
 - Delirium
 - Obtundation
 - Stupor
 - GCS ≤ 8 ⁽¹⁸⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, **Both:** ⁽³⁸⁾
 - Hypotension, **≥ One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Finding, **≥ One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽³⁹⁾
 - Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽⁴⁰⁾
- Mechanical ventilation
- NIPPV and, **≥ One:** ⁽³⁴⁾
 - Arterial $\text{Po}_2 \leq 39$ mmHg(5.2 kPa)
- Arterial or venous $\text{Pco}_2 \geq 60$ mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Oxygen, **Both:** ⁽¹¹⁾
 - HFNC ⁽³⁵⁾
 - Respiratory monitoring every 1-2h ⁽³⁶⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 1, One:** ⁽¹⁾
(Symptom or finding within 24h)
- **OBSERVATION, One:** ⁽³⁾
 - Diagnosis actual or suspected and addressed in a condition specific subset, **One:** ⁽⁴¹⁾
 - Abscess, cutaneous (**use Infection: Skin subset**)
 - *C. difficile* infection (**use Infection: GI/GYN subset**)
 - Cellulitis (**use Infection: Cellulitis subset**)

- COVID-19 (**use Infection: Covid-19 subset**)
- Gastroenteritis (**use Dehydration or Gastroenteritis subset**)
- Herpes zoster (**use Infection: Skin subset**)
- Meningitis (**use Infection: CNS subset**)
- Pelvic inflammatory disease (**use Infection: GI/GYN subset**)
- Pneumonia (**use Infection: Pneumonia subset**)
- Pyelonephritis or complex UTI (**use Infection: Pyelonephritis or Complicated UTI subset**)
- Wound infection (**use Infection: Skin subset**)
- Other infection actual or suspected, ≥ **One:** ⁽⁴²⁾
 - Dengue actual or suspected and, **Both:** ⁽⁴⁾
 - Finding, ≥ **One:** ⁽⁵⁾
 - Inadequate oral intake ⁽⁶⁾
 - Age > 65 yrs
 - Comorbidity, ≥ **One:**
 - Diabetes mellitus
 - Chronic renal disease
 - Hemolytic disease
 - Obesity
 - Emergency services remote
 - Unable to obtain follow-up care within 24h ⁽⁷⁾
 - Intervention, **All:** ⁽⁴³⁾
 - Hct monitoring at least 2x/24h ⁽⁴⁴⁾
 - IV fluid ⁽⁴⁵⁾
 - Vital sign monitoring at least 4x/24h
- Mononucleosis and, **Both:** ⁽⁸⁾
 - Finding, **Both:**
 - Impaired swallowing
 - Inadequate oral intake ⁽⁶⁾
 - IV fluid ⁽⁴⁶⁾
- Peritonsillar abscess actual suspected and, **Both:** ⁽⁹⁾
 - Drainage, surgical and, **One:**
 - Performed within last 24h
 - Anticipated within the next 24h ⁽⁴⁷⁾
 - Anti-infective
- Viral respiratory infection actual or suspected and, **All:**
 - Dyspnea
 - Oxygenation, ≥ **One:**
 - Arterial Po₂ 56-60 mmHg(7.4-8.0 kPa) and < baseline ⁽¹⁰⁾
 - O₂ sat 90-91%(0.90-0.91) and < baseline ⁽¹⁰⁾
 - Requiring supplemental oxygen ⁽¹¹⁾
- ACUTE, **One:** ⁽¹²⁾
 - Diagnosis actual or suspected and addressed in a condition specific subset, **One:** ⁽⁴¹⁾
 - Abscess cutaneous (**use Infection: Skin**)
 - Ascending cholangitis (**use Infection: GI/GYN subset**)
 - Brain abscess (**use Infection: CNS subset**)
 - Cellulitis (**use Infection: Cellulitis subset**)
 - *C. difficile* infection (**use Infection: GI/GYN subset**)
 - COVID-19 (**use Infection: Covid-19 subset**)
 - Diabetic foot ulcer (**use Infection: Skin subset**)
 - Diverticulitis (**use Infection: GI/GYN subset**)
 - Encephalitis (**use Infection: CNS subset**)
 - Endocarditis (**use Infection: Endocarditis subset**)
 - Gastroenteritis (**use Dehydration or Gastroenteritis subset**)

- Herpes zoster (**use Infection: Skin subset**)
- Intra-abdominal abscess (**use Infection: GI/GYN subset**)
- Meningitis (**use Infection: CNS subset**)
- Osteomyelitis (**use Infection: Musculoskeletal subset**)
- Pelvic abscess (**use Infection: GI/GYN subset**)
- Pelvic inflammatory disease (**use Infection: GI/GYN subset**)
- Pneumonia (**use Infection: Pneumonia subset**)
- Pyelonephritis or complex UTI (**use Infection: Pyelonephritis or Complex UTI subset**)
- Septic arthritis (**use Infection: Musculoskeletal subset**)
- Spontaneous bacterial peritonitis (**use Infection: GI/GYN subset**)
- Systemic infection unknown source (**use Infection: Sepsis subset**)
- Wound infection (**use Infection: Skin subset**)
- Other infection actual or suspected, ≥ **One:** ⁽⁴²⁾
 - Central venous catheter and, **Both:** ⁽¹³⁾
 - Finding, ≥ **One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
 - Outpatient blood culture positive
 - Purulent drainage at exit site ⁽¹⁵⁾
 - Anti-infective
 - Dengue actual or suspected and, **Both:** ⁽⁴⁾
 - Finding, ≥ **One:** ⁽¹⁶⁾
 - Abdominal pain
 - Ascites
 - Hct > 10%(0.10) increase from baseline
 - Hepatomegaly > 2 cm
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 ^(17, 48)
 - Mucosal bleeding
 - Platelet count < 100,000/cu.mm(100x10⁹/L)
 - Petechiae ⁽¹⁹⁾
 - Pleural effusion
 - Pregnancy
 - Vomiting, protracted ⁽²⁰⁾
 - Intervention, **All:** ⁽⁴⁹⁾
 - Hct monitoring at least 2x/24h ⁽⁴⁴⁾
 - IV fluid ⁽⁴⁵⁾
 - Vital sign monitoring at least 6x/24h
 - Fever of unknown origin and, **All:** ⁽²¹⁾
 - Temperature > 101.0°F(38.3°C) ≥ 3 wks
 - Outpatient work-up non-diagnostic ⁽²²⁾
 - Secondary evaluation ≤ 3d ⁽⁵⁰⁾
 - Intraocular infection and anti-infective ⁽²³⁾
- Laboratory result indicative of infection, **All:**
 - Test, ≥ **One:**
 - Positive culture for bacteria
 - Positive smear for bacteria
 - Positive antigen for bacteria
 - Positive latex agglutination for bacteria
 - Positive culture for fungi
 - Positive smear for fungi
 - Positive antigen for fungi
 - Positive latex agglutination for fungi
 - Positive culture for protozoa
 - Positive smear for protozoa

- Positive antigen for protozoa
- Positive latex agglutination for protozoa
- Source, ≥ **One**:
 - Biopsy, surgical (excludes skin) ⁽²⁴⁾
 - Blood
 - Bone marrow
 - Effluent smear ⁽²⁵⁾
- Anti-infective
- Mastoiditis and, **All**:
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
 - Confirmed by x-ray or CT
 - Erythema or pain over mastoid area
 - Anti-infective
- Neutropenia, **All**: ⁽²⁶⁾
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
 - ANC, **One**: ⁽²⁷⁾
 - < 500/cu.mm(500x10⁶/L)
 - < 1000/cu.mm(1000x10⁶/L) with documented decline ⁽²⁸⁾
 - Anti-infective
- New onset fever in the hospitalized patient and, **Both**: ^(51, 52)
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
 - Intervention ≤ 24h, ≥ **One**:
 - Culture pending
 - Drug-induced fever suspected and precipitating drug discontinued ⁽⁵³⁾
 - Imaging study
- Pregnancy and suspected infection and, **All**:
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
 - WBC, ≥ **One**:
 - > 18,000/cu.mm(18x10⁹/L)
 - > 15,000/cu.mm(15x10⁹/L) and bands > 7%(0.07)
 - Anti-infective
- Retropharyngeal abscess actual or suspected and, **Both**: ⁽²⁹⁾
 - Surgical evaluation and, **One**:
 - Performed within 24h
 - Scheduled within 24h
 - Anti-infective
- TB actual or suspected and, **Both**:
 - Negative airflow isolation
 - Finding, **One**:
 - Public health risk and anti-infective (includes PO) ⁽³⁰⁾
 - < 3 consecutive negative AFB sputum smears ⁽³¹⁾
 - Positive AFB sputum smear and anti-infective (includes PO)
- Typhoid fever actual or suspected and, **All**: ⁽³²⁾
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
 - Finding, ≥ **One**:
 - Abdominal distention
 - Abdominal pain
 - Inadequate oral intake ⁽⁶⁾
 - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14
 - Anti-infective
- Viral respiratory infection actual or suspected and, **All**:
 - Dyspnea
 - Oxygenation, ≥ **One**:

- Arterial $\text{Po}_2 < 56 \text{ mmHg}(7.4 \text{ kPa})$ and $< \text{baseline}^{(10)}$
- $\text{O}_2 \text{ sat} \leq 89\%(0.89)$ and $< \text{baseline}^{(10)}$
- Requiring supplemental oxygen⁽¹¹⁾

● **INTERMEDIATE, One:**⁽³³⁾

- Diagnosis actual or suspected, and addressed in a condition specific subset, **One:**⁽⁴¹⁾
 - Ascending cholangitis (**use Infection: GI/GYN subset**)
 - Brain abscess (**use Infection: CNS subset**)
 - *C. difficile* infection (**use Infection: GI/GYN subset**)
 - COVID-19 (**use Infection: Covid-19 subset**)
 - Diverticulitis (**use Infection: GI/GYN subset**)
 - Encephalitis (**use Infection: CNS subset**)
 - Endocarditis (**use Infection: Endocarditis subset**)
 - Gastroenteritis (**use Dehydration or Gastroenteritis subset**)
 - Intra-abdominal abscess (**use Infection: GI/GYN subset**)
 - Meningitis (**use Infection: CNS subset**)
 - Pelvic abscess (**use Infection: GI/GYN subset**)
 - Pelvic inflammatory disease (**use Infection: GI/GYN subset**)
 - Pneumonia (**use Infection: Pneumonia subset**)
 - Pyelonephritis or complex UTI (**use Infection: Pyelonephritis or Complex UTI subset**)
 - Sepsis (**use Infection: Sepsis subset**)
 - Spontaneous bacterial peritonitis (**use Infection: GI/GYN subset**)
- Other infection actual or suspected, $\geq \text{One}$:
 - NIPPV⁽³⁴⁾
- Oxygen, $\geq \text{One}$:⁽¹¹⁾
 - HFNC and respiratory monitoring every 3-4h^(35, 36)
 - Oxygen $\geq 40\%(0.40) \leq 2\text{d}$

● **CRITICAL, One:**⁽³⁷⁾

- Diagnosis actual or suspected and addressed in a condition specific subset, **One:**⁽⁴¹⁾
 - Ascending cholangitis (**use Infection: GI/GYN subset**)
 - Brain abscess (**use Infection: CNS subset**)
 - *C. difficile* infection (**use Infection: GI/GYN subset**)
 - COVID-19 (**use Infection: Covid-19 subset**)
 - Diverticulitis (**use Infection: GI/GYN subset**)
 - Encephalitis (**use Infection: CNS subset**)
 - Endocarditis (**use Infection: Endocarditis subset**)
 - Gastroenteritis (**use Dehydration or Gastroenteritis subset**)
 - Intra-abdominal abscess (**use Infection: GI/GYN subset**)
 - Meningitis (**use Infection: CNS subset**)
 - Pelvic abscess (**use Infection: GI/GYN subset**)
 - Pelvic inflammatory disease (**use Infection: GI/GYN subset**)
 - Pneumonia (**use Infection: Pneumonia subset**)
 - Pyelonephritis or complex UTI (**use Infection: Pyelonephritis or Complex UTI subset**)
 - Sepsis (**use Infection: Sepsis subset**)
 - Spontaneous bacterial peritonitis (**use Infection: GI/GYN subset**)
- Other infection actual or suspected, $\geq \text{One}$:⁽⁴²⁾
 - Epiglottitis actual or suspected and, **All:**
 - Risk of airway compromise
 - Finding, $\geq \text{One}$:
 - Confirmed by x-ray
 - Difficulty swallowing
 - Drooling
 - Oximetry
- Peritonsillar abscess actual or suspected and, **All:**⁽⁹⁾

- Risk of airway compromise
- Finding, ≥ **One**:
 - Confirmed by x-ray or CT
 - Difficulty swallowing
 - Drooling
- Intervention, **Both**:
 - Oximetry
 - Surgical evaluation performed within 24h
- Retropharyngeal abscess actual or suspected and, **All**: ⁽²⁹⁾
 - Risk of airway compromise
- Finding, ≥ **One**:
 - Confirmed by x-ray or CT
 - Difficulty swallowing
 - Drooling
- Intervention, **Both**:
 - Oximetry
 - Surgical evaluation and, **One**:
 - Performed within 24h
 - Scheduled within 24h
- Typhoid fever actual or suspected and, ≥ **One**: ⁽³²⁾
 - Gastrointestinal bleeding and, **One**:
 - Blood product transfusion ≤ 24h
 - Mental status change, **Both**: ⁽¹⁷⁾
 - Finding, ≥ **One**:
 - Coma
 - Delirium
 - Obtundation
 - Stupor
 - GCS ≤ 8 ⁽¹⁸⁾
 - Neurological assessment every 1-2h ⁽⁵⁴⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both**: ⁽³⁸⁾
 - Hypotension, ≥ **One**:
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Finding, ≥ **One**:
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽³⁹⁾
 - Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽⁴⁰⁾
- Mechanical ventilation
- NIPPV and, ≥ **One**: ⁽³⁴⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One**:
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Oxygen, **Both**: ⁽¹¹⁾
 - HFNC ⁽³⁵⁾
 - Respiratory monitoring every 1-2h ⁽³⁶⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- Episode Day 2, **One**: ⁽⁵⁵⁾
 - OBSERVATION, **One**:
 - Responder, discharge expected today if clinically stable last 12h, ≥ **One**: ⁽⁵⁶⁾
 - Dengue and, **All**:

- Lab values within acceptable limits ⁽⁵⁷⁾
- Tolerating PO ⁽⁵⁸⁾
- Urine output within acceptable limits ⁽⁵⁷⁾
- Mononucleosis and, **Both:**
 - Airway patent
 - Tolerating PO ⁽⁵⁹⁾
- Peritonsillar abscess and, **Both:** ⁽⁹⁾
 - Airway patent
 - Tolerating PO ⁽⁵⁹⁾
- Viral respiratory infection and, **≥ One:**
 - Arterial Po₂ ≥ 60 mmHg(7.9 kPa) or within acceptable limits ⁽⁵⁷⁾
 - O₂ sat within acceptable limits ⁽⁵⁷⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, **≥ One:**
 - Dengue and, **All:**
 - Hct monitoring at least 2x/24h ⁽⁴⁴⁾
 - Inadequate oral intake ⁽⁶⁾
 - IV fluid ⁽⁴⁵⁾
 - Vital sign monitoring at least 4x/24h
 - Mononucleosis and, **Both:**
 - Finding, **Both:**
 - Impaired swallowing
 - Inadequate oral intake ⁽⁶⁰⁾
 - IV fluid ⁽⁴⁶⁾
 - Peritonsillar abscess and, **All:** ⁽⁹⁾
 - Anti-infective ⁽⁶¹⁾
 - Drainage, surgical, performed within last 24h
 - IV fluid ⁽⁴⁶⁾
 - Viral respiratory infection and, **All:**
 - Dyspnea
 - Oxygenation, **≥ One:**
 - Arterial Po₂ 56-60 mmHg(7.4-8.0 kPa) and < baseline ⁽¹⁰⁾
 - O₂ sat 90-91%(0.90-0.91) and < baseline ⁽¹⁰⁾
 - Requiring supplemental oxygen ⁽¹¹⁾
- **ACUTE, One:**
 - **Early responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁵⁶⁾
 - Afebrile
 - Clinical stability, **≥ One:**
 - Infection actual or suspected and, **One:**
 - Resolving and, **≥ One:** ⁽⁶²⁾
 - Anti-infective regimen established and tolerated
 - Symptoms resolving
 - Ruled out
 - Dengue and, **All:**
 - Lab values within acceptable limits ⁽⁵⁷⁾
 - Pain controlled or manageable ⁽⁶³⁾
 - Tolerating PO ⁽⁵⁹⁾
 - Urine output within acceptable limits ⁽⁵⁷⁾
 - Intraocular infection and, **Both:**
 - Infection resolving ⁽⁶²⁾
 - Vision normal, improving, or stable
 - Neutropenia and ANC ≥ 500/cu.mm(500x10⁶/L) ⁽²⁷⁾
 - Viral respiratory infection and O₂ sat within acceptable limits ⁽⁵⁷⁾
 - Tuberculosis and, **Both:**

- AFB sputum smear negative
- Safe for discharge ⁽⁶⁴⁾
- **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁶⁵⁾
 - **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - Adverse drug reaction resolving
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽⁵⁷⁾
 - Tolerating PO ⁽⁵⁹⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Central venous catheter infection and anti-infection ≤ 5d since initiation ^(13, 61)
 - Dengue and Hct monitoring at least 2x/24h ^(4, 44)
 - **Epiglottitis without airway compromise and, ≥ One:**
 - Anti-infective ≤ 5d since initiation ⁽⁶¹⁾
 - Inadequate oral intake and IV fluid ≤ 4d ^(46, 66)
 - Fever of unknown origin and secondary evaluation ≤ 3d since admission ⁽⁵⁰⁾
 - Intraocular infection and anti-infective ≤ 5d since initiation ^(23, 61)
 - Mastoiditis and anti-infective ≤ 5d since initiation ⁽⁶¹⁾
 - **Neutropenia, All:** ⁽²⁶⁾
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
 - **ANC, One:** ⁽²⁷⁾
 - < 500/cu.mm(500x10⁶/L)
 - < 1000/cu.mm(1000x10⁶/L) with documented decline ⁽²⁸⁾
 - Anti-infective
 - **Pregnancy and, All:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
 - WBC ≥ 12,000/cu.mm(12.0x10⁹/L)
 - Cultures pending
 - Anti-infective
 - **Retropharyngeal abscess and, All:**
 - Anti-infective ≤ 5d since initiation ⁽⁶¹⁾
 - **Drainage surgical and, One:**
 - Performed within last 24h
 - Scheduled within the next 48h
 - IV fluid ⁽⁴⁶⁾
 - **TB actual or suspected and, Both:**
 - Negative airflow isolation
 - **Finding, One:**
 - TB suspected and < 3 consecutive negative AFB sputum smears ⁽³¹⁾
 - **TB confirmed and not safe for discharge and, Both:** ⁽⁶⁴⁾
 - Public health risk ⁽³⁰⁾
 - Anti-infective (includes PO)
 - **TB confirmed by positive AFB smear and, Both:**
 - < 3 consecutive negative AFB sputum smears ⁽³¹⁾
 - Anti-infective (includes PO)
 - **Typhoid fever and, One:**
 - Bowel perforation (**use General Surgical subset**)
 - Anti-infective
 - **Viral respiratory infection actual or suspected and, All:**
 - Dyspnea
 - **Oxygenation, ≥ One:**
 - Arterial Po₂ < 56 mmHg(7.4 kPa) and < baseline ⁽¹⁰⁾
 - O₂ sat ≤ 89%(0.89) and < baseline ⁽¹⁰⁾
 - Requiring supplemental oxygen ⁽¹¹⁾

- Adverse drug reaction requiring change in medication $\leq 24\text{h}$ ⁽⁶⁷⁾
- Blood glucose $\leq 24\text{h}$, **All:** ⁽⁶⁸⁾
 - Finding, $\geq$ **One:**
 - $> 250\text{ mg/dL}(13.9\text{ mmol/L})$
 - $< 70\text{ mg/dL}(3.9\text{ mmol/L})$
 - Blood glucose monitoring at least 4x/24h
 - Insulin adjustment and, **One:** ^(68, 69)
 - IV $\geq 2\text{x}/24\text{h}$
 - SC $\geq 4\text{x}/24\text{h}$
 - Post Critical care $\leq 24\text{h}$
- **INTERMEDIATE, $\geq$ One:**
 - NIPPV ⁽³⁴⁾
 - Oxygen, $\geq$ **One:** ⁽¹¹⁾
 - HFNC and respiratory monitoring every 3-4h ^(35, 36)
 - Oxygen $\geq 40\%(0.40) \leq 2\text{d}$
- **CRITICAL, $\geq$ One:**
 - Epiglottitis and, **Both:**
 - Risk of airway compromise
 - Oximetry
 - Peritonsillar abscess and, **All:**
 - Risk of airway compromise
 - Finding, $\geq$ **One:**
 - Confirmed by x-ray or CT
 - Difficulty swallowing
 - Drooling
 - Intervention, **Both:**
 - Drainage surgical and, **One:**
 - Performed within last 24h
 - Scheduled within the next 48h
 - Oximetry
 - Retropharyngeal abscess and, **All:**
 - Risk of airway compromise
 - Finding, $\geq$ **One:**
 - Confirmed by x-ray or CT
 - Difficulty swallowing
 - Drooling
 - Intervention, **Both:**
 - Drainage surgical and, **One:**
 - Performed within last 24h
 - Scheduled within the next 48h
 - Oximetry
 - Typhoid fever and, $\geq$ **One:** ⁽³²⁾
 - Bowel perforation (**use General Surgical subset**)
 - Gastrointestinal bleeding and blood product transfusion $\leq 24\text{h}$
 - Mental status change, **Both:** ⁽¹⁷⁾
 - Finding, $\geq$ **One:**
 - Coma
 - Delirium
 - Obtundation
 - Stupor
 - GCS ≤ 8 ⁽¹⁸⁾
 - Neurological assessment every 1-2h ⁽⁵⁴⁾
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation $\leq 24\text{h}$

and, **Both:** ⁽³⁸⁾

- **Hypotension, ≥ One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline

- **Finding, ≥ One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽³⁹⁾
 - Urine output < 0.5 mL/kg/h

– Hemodynamic monitoring, invasive ⁽⁴⁰⁾

– Mechanical ventilation

- **NIPPV and, ≥ One:** ⁽³⁴⁾

- FiO₂ > 50%(0.50)
- Respiratory intervention every 1-2h ⁽⁷⁰⁾

– HFNC and respiratory monitoring every 1-2h ^(11, 35, 36)

– Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

- **Episode Day 3, One:** ⁽⁷¹⁾

- **OBSERVATION, One:**

- **Responder**, discharge expected today if clinically stable last 12h, ≥ **One:** ⁽⁵⁶⁾

- **Dengue and, All:**
 - Lab values within acceptable limits ⁽⁵⁷⁾
 - Tolerating PO ⁽⁵⁸⁾
 - Urine output within acceptable limits ⁽⁵⁷⁾

- **Mononucleosis and, Both:**

- Airway patent
- Tolerating PO ⁽⁵⁹⁾

- **Peritonsillar abscess and, Both:** ⁽⁹⁾

- Airway patent
- Tolerating PO ⁽⁵⁹⁾

- **Viral respiratory infection and, ≥ One:**

- Arterial Po₂ within acceptable limits ⁽⁵⁷⁾
- O₂ sat within acceptable limits ⁽⁵⁷⁾

- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**

– For current condition use Episode Day 3 at a higher level of care

– For a condition other than current condition use appropriate subset (Episode Day 1) based on clinical findings

– Refer for secondary review

- **ACUTE, One:**

- **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁵⁶⁾

– Afebrile

- **Clinical stability, ≥ One:**

- **Infection and, One:**

- **Resolving and, ≥ One:** ⁽⁶²⁾

- Anti-infective regimen established and tolerated
- Symptoms resolving

– Ruled out

- **Dengue and, All:**

- Lab values within acceptable limits ⁽⁵⁷⁾
- Pain controlled or manageable ⁽⁶³⁾
- Tolerating PO ⁽⁵⁹⁾
- Urine output within acceptable limits ⁽⁵⁷⁾

- **Intraocular infection and, Both:**

- Infection resolving ⁽⁶²⁾
- Vision normal, improving, or stable

- Neutropenia and ANC $\geq 500/\text{cu.mm}(500 \times 10^6/\text{L})$ ⁽²⁷⁾
- TB and, **Both:**
 - AFB sputum smear negative
 - Safe for discharge ⁽⁶⁴⁾
 - Viral respiratory infection and O₂ sat within acceptable limits ⁽⁵⁷⁾
- Complication or comorbidity, **One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁶⁵⁾
- Complication or comorbidity and clinically stable for discharge, $\geq$ **One:**
 - Adverse drug reaction resolving
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽⁵⁷⁾
- *C. difficile* infection and, **Both:** ⁽⁷²⁾
 - Decrease in frequency and severity of diarrhea
 - Tolerating PO ⁽⁵⁸⁾
 - Tolerating PO ⁽⁵⁹⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - Central venous catheter infection and anti-infection $\leq 5\text{d}$ since initiation ^(13, 61)
 - Dengue and Hct monitoring at least 2x/24h ^(4, 44)
- Epiglottitis without airway compromise and, $\geq$ **One:**
 - Anti-infective $\leq 5\text{d}$ since initiation ⁽⁶¹⁾
 - Inadequate oral intake and IV fluid $\leq 4\text{d}$ ^(46, 66)
- Fever of unknown origin and secondary evaluation $\leq 3\text{d}$ since admission ⁽⁵⁰⁾
- Intraocular infection and anti-infective $\leq 5\text{d}$ since initiation ^(23, 61)
- Mastoiditis and anti-infective $\leq 5\text{d}$ since initiation ⁽⁶¹⁾
- Mononucleosis and, **Both:**
 - Inadequate oral intake
 - IV fluid $\leq 4\text{d}$ ⁽⁴⁶⁾
- Neutropenia, **All:** ⁽²⁶⁾
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ ⁽¹⁴⁾
- ANC, **One:** ⁽²⁷⁾
 - $< 500/\text{cu.mm}(500 \times 10^6/\text{L})$
 - $< 1000/\text{cu.mm}(1000 \times 10^6/\text{L})$ with documented decline ⁽²⁸⁾
- Anti-infective
- Peritonsillar abscess and, **Both:** ⁽⁹⁾
 - Anti-infective $\leq 5\text{d}$ since initiation ⁽⁶¹⁾
 - IV fluid ⁽⁴⁶⁾
- Pregnancy and, **All:**
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ ⁽¹⁴⁾
 - WBC $\geq 12,000/\text{cu.mm}(12.0 \times 10^9/\text{L})$
 - Cultures pending
 - Anti-infective
- Retropharyngeal abscess and, **All:**
 - Anti-infective $\leq 5\text{d}$ since initiation ⁽⁶¹⁾
- Drainage surgical and, **One:**
 - Performed within last 24h
 - Scheduled within the next 24h
- IV fluid ⁽⁴⁶⁾
- TB actual or suspected and, **Both:**
 - Negative airflow isolation
- Finding, **One:**
 - TB suspected and < 3 consecutive negative AFB sputum smears ⁽³¹⁾
- TB confirmed and not safe for discharge and, **Both:** ⁽⁶⁴⁾
 - Public health risk ⁽³⁰⁾
 - Anti-infective (includes PO)

- TB confirmed by positive AFB smear and, **Both:**
 - < 3 consecutive negative AFB sputum smears ⁽³¹⁾
 - Anti-infective (includes PO)
- Typhoid fever and, **One:**
 - Bowel perforation (**use General Surgical subset**)
 - Anti-infective
- Viral respiratory infection and, **Both:**
 - O₂ sat ≤ 91%(0.91) and < baseline ^(10, 11)
 - Requiring supplemental oxygen
- Adverse drug reaction requiring change in medication ≤ 24h ⁽⁶⁷⁾
- Blood glucose ≤ 24h, **All:** ⁽⁶⁸⁾
 - Finding, ≥ **One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring at least 4x/24h
- Insulin adjustment and, **One:** ^(68, 69)
 - IV ≥ 2x/24h
 - SC ≥ 4x/24h
- *C. difficile* infection, actual or suspected and, **Both:** ⁽⁷³⁾
 - Anti-infective (includes PO) ≤ 5d ^(74, 75)
 - Inadequate oral intake ⁽⁷⁶⁾
- Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - NIPPV ⁽³⁴⁾
- Oxygen, ≥ **One:** ⁽¹¹⁾
 - HFNC and respiratory monitoring every 3-4h ^(35, 36)
 - Oxygen ≥ 40%(0.40) ≤ 2d
- **CRITICAL, ≥ One:**
 - Epiglottitis and, **Both:**
 - Risk of airway compromise
 - Oximetry
 - Peritonsillar abscess and, **All:**
 - Risk of airway compromise
 - Finding, ≥ **One:**
 - Confirmed by x-ray or CT
 - Difficulty swallowing
 - Drooling
 - Oximetry
 - Retropharyngeal abscess and, **All:**
 - Risk of airway compromise
 - Finding, ≥ **One:**
 - Confirmed by x-ray or CT
 - Difficulty swallowing
 - Drooling
 - Oximetry
 - Typhoid fever and, ≥ **One:** ⁽³²⁾
 - Bowel perforation (**use General Surgical subset**)
 - Gastrointestinal bleeding and blood product transfusion ≤ 24h
 - Mental status change, **Both:** ⁽¹⁷⁾
 - Finding, ≥ **One:**
 - Coma
 - Delirium
 - Obtundation

- Stupor
- GCS ≤ 8 ⁽¹⁸⁾
- Neurological assessment every 1-2h ⁽⁵⁴⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, **Both:** ⁽³⁸⁾
 - Hypotension, $\geq$ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Finding, $\geq$ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽³⁹⁾
 - Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽⁴⁰⁾
- Mechanical ventilation
- NIPPV and, $\geq$ **One:** ⁽³⁴⁾
 - FiO₂ $> 50\%$ (0.50)
 - Respiratory intervention every 1-2h ⁽⁷⁰⁾
- HFNC and respiratory monitoring every 1-2h ^(11, 35, 36)
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- Episode Day 4-6, **One:**
 - ACUTE, **One:**
 - Responder, discharge expected today if clinically stable last 12h, **All:** ⁽⁵⁶⁾
 - Afebrile
 - Clinical stability, $\geq$ **One:**
 - Infection resolving, $\geq$ **One:** ⁽⁶²⁾
 - Anti-infective regimen established and tolerated
 - Symptoms resolving
 - Dengue and, **All:**
 - Lab values within acceptable limits ⁽⁵⁷⁾
 - Pain controlled or manageable ⁽⁶³⁾
 - Tolerating PO ⁽⁵⁹⁾
 - Urine output within acceptable limits ⁽⁵⁷⁾
 - Epiglottitis and, **Both:**
 - Airway patent
 - Tolerating PO ⁽⁵⁹⁾
 - Intraocular infection and, **Both:**
 - Infection resolving ⁽⁶²⁾
 - Vision normal, improving, or stable
 - Mastoiditis and, **All:**
 - Pain controlled or manageable ⁽⁶³⁾
 - Swelling decreased
 - Anti-infective regimen established and tolerated
 - Mononucleosis and, **Both:**
 - Airway patent
 - Tolerating PO ⁽⁵⁹⁾
 - Neutropenia and ANC ≥ 500 /cu.mm(500x10⁶/L) ⁽²⁷⁾
 - Peritonsillar abscess resolving and, **Both:**
 - Airway patent
 - Tolerating PO ⁽⁵⁹⁾
 - Retropharyngeal abscess resolving and, **Both:**
 - Airway patent
 - Tolerating PO ⁽⁵⁹⁾
 - TB and, **Both:**
 - AFB sputum smear negative

- Safe for discharge ⁽⁶⁴⁾
- Typhoid fever and, **All:** ⁽⁹⁾
 - Lab values within acceptable limits ⁽⁵⁷⁾
 - Pain controlled or manageable ⁽⁶³⁾
 - Neurological stability ⁽⁷⁷⁾
 - Tolerating PO ⁽⁵⁹⁾
 - Viral respiratory infection and O₂ sat within acceptable limits ⁽⁵⁷⁾
- Complication or comorbidity, **One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁶⁵⁾
 - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
 - Adverse drug reaction resolving
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽⁵⁷⁾
 - *C. difficile* infection and, **Both:** ⁽⁷²⁾
 - Decrease in frequency and severity of diarrhea
 - Tolerating PO ⁽⁵⁸⁾
 - Tolerating PO ⁽⁵⁹⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Central venous catheter infection and anti-infection ≤ 5d since initiation ^(13, 61)
 - Dengue and Hct monitoring at least 2x/24h ^(4, 44)
 - Epiglottitis without airway compromise and, ≥ **One:**
 - Anti-infective ≤ 5d since initiation ⁽⁶¹⁾
 - Inadequate oral intake and IV fluid ≤ 4d ^(46, 66)
 - Intraocular infection and anti-infective ≤ 5d since initiation ^(23, 61)
 - Mastoiditis and anti-infective ≤ 5d since initiation ⁽⁶¹⁾
 - Mononucleosis and, **Both:**
 - Inadequate oral intake
 - IV fluid ≤ 4d ⁽⁴⁶⁾
 - Neutropenia, **All:** ⁽²⁶⁾
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
 - ANC, **One:** ⁽²⁷⁾
 - < 500/cu.mm(500x10⁶/L)
 - < 1000/cu.mm(1000x10⁶/L) with documented decline ⁽²⁸⁾
 - Anti-infective
 - Peritonsillar abscess and, **Both:** ⁽⁹⁾
 - Anti-infective ≤ 5d since initiation ⁽⁶¹⁾
 - IV fluid ⁽⁴⁶⁾
 - Pregnancy and, **All:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹⁴⁾
 - WBC ≥ 12,000/cu.mm(12.0x10⁹/L)
 - Anti-infective
 - Retropharyngeal abscess and, **Both:**
 - Anti-infective ≤ 5d since initiation ⁽⁶¹⁾
 - IV fluid ⁽⁴⁶⁾
 - TB actual or suspected and, **Both:**
 - Negative airflow isolation
 - Finding, **One:**
 - TB suspected and < 3 consecutive negative AFB sputum smears ⁽³¹⁾
 - TB confirmed and not safe for discharge and, **Both:** ⁽⁶⁴⁾
 - Public health risk ⁽³⁰⁾
 - Anti-infective (includes PO)
 - TB confirmed by positive AFB smear and, **Both:**
 - < 3 consecutive negative AFB sputum smears ⁽³¹⁾
 - Anti-infective (includes PO)

- Typhoid fever and, **One:**
 - Bowel perforation (use **General Surgical subset**)
 - Anti-infective
- Viral respiratory infection and, **Both:**
 - O₂ sat ≤ 91%(0.91) and < baseline ^(10, 11)
 - Requiring supplemental oxygen
- Adverse drug reaction requiring change in medication ≤ 24h ⁽⁶⁷⁾
- Blood glucose ≤ 24h, **All:** ⁽⁶⁸⁾
 - Finding, ≥ **One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring at least 4x/24h
 - Insulin adjustment and, **One:** ^(68, 69)
 - IV ≥ 2x/24h
 - SC ≥ 4x/24h
- *C. difficile* infection, actual or suspected and, **Both:** ⁽⁷³⁾
 - Anti-infective (includes PO) ≤ 5d ^(74, 75)
 - Inadequate oral intake ⁽⁷⁶⁾
- Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - NIPPV ⁽³⁴⁾
 - Oxygen, ≥ **One:** ⁽¹¹⁾
 - HFNC and respiratory monitoring every 3-4h ^(35, 36)
 - Oxygen ≥ 40%(0.40) ≤ 2d
- **CRITICAL, ≥ One:**
 - Epiglottitis and, **Both:**
 - Risk of airway compromise
 - Oximetry
 - Peritonsillar abscess and, **All:**
 - Risk of airway compromise
 - Finding, ≥ **One:**
 - Confirmed by x-ray or CT
 - Difficulty swallowing
 - Drooling
 - Oximetry
 - Retropharyngeal abscess and, **All:**
 - Risk of airway compromise
 - Finding, ≥ **One:**
 - Confirmed by x-ray or CT
 - Difficulty swallowing
 - Drooling
 - Oximetry
 - Typhoid fever and, ≥ **One:** ⁽³²⁾
 - Bowel perforation (use **General Surgical subset**)
 - Gastrointestinal bleeding and blood product transfusion ≤ 24h
 - Mental status change, **Both:** ⁽¹⁷⁾
 - Finding, ≥ **One:**
 - Coma
 - Delirium
 - Obtundation
 - Stupor
 - GCS ≤ 8 ⁽¹⁸⁾
 - Neurological assessment every 1-2h ⁽⁵⁴⁾

- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, **Both:** ⁽³⁸⁾
 - Hypotension, $\geq$ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Finding, $\geq$ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽³⁹⁾
 - Urine output < 0.5 mL/kg/h
 - Hemodynamic monitoring, invasive ⁽⁴⁰⁾
 - Mechanical ventilation
 - NIPPV and, $\geq$ **One:** ⁽³⁴⁾
 - FiO₂ $> 50\%$ (0.50)
 - Respiratory intervention every 1-2h ⁽⁷⁰⁾
 - HFNC and respiratory monitoring every 1-2h ^(11, 35, 36)
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- Episode Day 7, **One:**
 - ACUTE, **One:**
 - Responder, discharge expected today if clinically stable last 12h, **All:** ⁽⁵⁶⁾
 - Afebrile
 - Clinical stability, $\geq$ **One:**
 - Infection resolving, $\geq$ **One:** ⁽⁶²⁾
 - Anti-infective regimen established and tolerated
 - Symptoms resolving
 - Dengue and, **All:**
 - Lab values within acceptable limits ⁽⁵⁷⁾
 - Pain controlled or manageable ⁽⁶³⁾
 - Tolerating PO ⁽⁵⁹⁾
 - Urine output within acceptable limits ⁽⁵⁷⁾
 - Epiglottitis and, **Both:**
 - Airway patent
 - Tolerating PO ⁽⁵⁹⁾
 - Intraocular infection and, **Both:**
 - Infection resolving ⁽⁶²⁾
 - Vision normal, improving, or stable
 - Mastoiditis and, **All:**
 - Pain controlled or manageable ⁽⁶³⁾
 - Swelling decreased
 - Anti-infective regimen established and tolerated
 - Mononucleosis and, **Both:**
 - Airway patent
 - Tolerating PO ⁽⁵⁹⁾
 - Neutropenia and ANC $\geq 500/\text{cu.mm}(500 \times 10^6/\text{L})$ ⁽²⁷⁾
 - Peritonsillar abscess resolving and, **Both:**
 - Airway patent
 - Tolerating PO ⁽⁵⁹⁾
 - Retropharyngeal abscess resolving and, **Both:**
 - Airway patent
 - Tolerating PO ⁽⁵⁹⁾
 - TB and, **Both:**
 - AFB sputum smear negative
 - Safe for discharge ⁽⁶⁴⁾
 - Typhoid fever and, **All:**
 - Lab values within acceptable limits ⁽⁵⁷⁾

- Pain controlled or manageable ⁽⁶³⁾
- Neurological stability ⁽⁷⁷⁾
- Tolerating PO ⁽⁵⁹⁾
- Viral respiratory infection and O₂ sat within acceptable limits ⁽⁵⁷⁾
- **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁶⁵⁾
- **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - Adverse drug reaction resolving
 - Blood glucose 70-250 mg/dL(3.9-13.9 mmol/L) or within acceptable limits ⁽⁵⁷⁾
- ***C. difficile* infection and, Both: ⁽⁷²⁾**
 - Decrease in frequency and severity of diarrhea
 - Tolerating PO ⁽⁵⁸⁾
 - Tolerating PO ⁽⁵⁹⁾
- **Non-responder, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, One:**
 - Use Extended Stay subset
 - Refer for secondary review
- **INTERMEDIATE, One:**
 - **Non-responder, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, One:**
 - Use Extended Stay subset
 - Refer for secondary review
- **CRITICAL, One:**
 - **Non-responder, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, One:**
 - Use Extended Stay subset
 - Refer for secondary review

Discharge, Level of Care, ≥ One: ⁽⁷⁸⁾

- **HOME, Both:**
 - **Level of care appropriateness, Both:**
 - Home environment safe and accessible ⁽⁷⁹⁾
 - Patient or caregiver demonstrates ability to manage care
 - **Discharge planning, All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁸⁰⁾
 - Medication reconciliation ⁽⁸¹⁾
 - Follow-up care planned ⁽⁸²⁾
 - Identify and address transportation needs
- **HOME CARE, Both:**
 - **Level of care appropriateness, All:**
 - Home environment safe and accessible ⁽⁷⁹⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(83, 84)
 - **Skilled services, ≥ One: ⁽⁸⁵⁾**
 - Skilled nursing ⁽⁸⁶⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁸⁷⁾
 - **Discharge planning, All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁸⁰⁾
 - Medication reconciliation ⁽⁸¹⁾
 - Follow-up care planned ⁽⁸²⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**

- Level of care appropriateness, **One:**
 - Support systems available ⁽⁸⁸⁾
- Skilled treatment, **≥ One:**
 - Clinical assessment ⁽⁸⁹⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁹⁰⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 1\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment $1\text{-}2\text{x}/24\text{h}$ ⁽⁸⁶⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(91, 92)
 - Functional impairments requiring at least supervision ⁽⁹³⁾
 - Able to tolerate 1h/d of skilled therapy $\geq 5\text{d/wk}$
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 2\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment $4\text{h}/24\text{h}$ ⁽⁸⁶⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(91, 92)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁹³⁾
 - Able to tolerate $2\text{-}3\text{h/d}$ of skilled therapy $\geq 5\text{d/wk}$ and ≥ 1 therapy discipline
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(91, 92)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁹³⁾
 - Able to tolerate $2\text{-}3\text{h/d}$ of skilled therapy $\geq 5\text{d/wk}$ and ≥ 2 therapy disciplines
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - Level of care appropriateness, **Both:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$
 - Services, **All:**

- Goal directed therapy with rehabilitation potential ^(91, 92)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁹³⁾
- Able to tolerate ≥ 15 h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁹⁴⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁹⁵⁾
 - Ventilator dependent ≥ 6 h/d and weaning planned ^(96, 97)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁹⁸⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸¹⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
094 BACTERIAL AND TUBERCULOUS INFECTIONS OF NERVOUS SYSTEM WITH MCC	n/a	8.6	CMS GMLOS
095 BACTERIAL AND TUBERCULOUS INFECTIONS OF NERVOUS SYSTEM WITH CC	n/a	5.9	CMS GMLOS
096 BACTERIAL AND TUBERCULOUS INFECTIONS OF NERVOUS SYSTEM WITHOUT CC/MCC	n/a	4.5	CMS GMLOS
121 ACUTE MAJOR EYE INFECTIONS WITH CC/MCC	n/a	4.4	CMS GMLOS
122 ACUTE MAJOR EYE INFECTIONS WITHOUT CC/MCC	n/a	3	CMS GMLOS
152 OTITIS MEDIA AND URI WITH MCC	n/a	3.2	CMS GMLOS
153 OTITIS MEDIA AND URI WITHOUT MCC	n/a	2.3	CMS GMLOS
177 RESPIRATORY INFECTIONS AND INFLAMMATIONS WITH MCC	n/a	5	CMS GMLOS
178 RESPIRATORY INFECTIONS AND INFLAMMATIONS WITH CC	n/a	3.3	CMS GMLOS
179 RESPIRATORY INFECTIONS AND INFLAMMATIONS WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
314 OTHER CIRCULATORY SYSTEM DIAGNOSES WITH MCC	n/a	5.1	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
315 OTHER CIRCULATORY SYSTEM DIAGNOSES WITH CC	n/a	2.7	CMS GMLOS
316 OTHER CIRCULATORY SYSTEM DIAGNOSES WITHOUT CC/MCC	n/a	1.8	CMS GMLOS
808 MAJOR HEMATOLOGICAL AND IMMUNOLOGICAL DIAGNOSES EXCEPT SICKLE CELL CRISIS AND COAGULATION DISORDERS WITH MCC	n/a	5.3	CMS GMLOS
809 MAJOR HEMATOLOGICAL AND IMMUNOLOGICAL DIAGNOSES EXCEPT SICKLE CELL CRISIS AND COAGULATION DISORDERS WITH CC	n/a	3.4	CMS GMLOS
810 MAJOR HEMATOLOGICAL AND IMMUNOLOGICAL DIAGNOSES EXCEPT SICKLE CELL CRISIS AND COAGULATION DISORDERS WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
811 RED BLOOD CELL DISORDERS WITH MCC	n/a	3.8	CMS GMLOS
812 RED BLOOD CELL DISORDERS WITHOUT MCC	n/a	2.8	CMS GMLOS
831 OTHER ANTEPARTUM DIAGNOSES WITHOUT O.R. PROCEDURES WITH MCC	n/a	3.3	CMS GMLOS
832 OTHER ANTEPARTUM DIAGNOSES WITHOUT O.R. PROCEDURES WITH CC	n/a	2.5	CMS GMLOS
833 OTHER ANTEPARTUM DIAGNOSES WITHOUT O.R. PROCEDURES WITHOUT CC/MCC	n/a	1.9	CMS GMLOS
865 VIRAL ILLNESS WITH MCC	n/a	4.3	CMS GMLOS
866 VIRAL ILLNESS WITHOUT MCC	n/a	2.8	CMS GMLOS
867 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITH MCC	n/a	5.5	CMS GMLOS
868 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITH CC	n/a	3.7	CMS GMLOS
869 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
CENTRAL VENOUS CATHETER INFECTION	2%	8.4	InterQual
EPIGLOTTITIS	23%	2.8	InterQual
INFECTION IN PREGNANCY	47%	2.3	InterQual
MASTOIDITIS	39%	3.1	InterQual
MONONUCLEOSIS	45%	2.6	InterQual
PERITONSILLAR ABSCESS	54%	1.8	InterQual

Condition or Procedure	% Pd Obs	LOS (days)	Type
RETROPHARYNGEAL ABSCESS	18%	3.2	InterQual
TUBERCULOSIS	17%	8	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

Dengue is a viral illness primarily transmitted to humans through the bite of infected mosquitoes, and is mainly limited to tropical and subtropical climates. Risk of transmission is increased through travel to locations where dengue is endemic.

The clinical presentation of dengue varies widely, ranging from a mild virus to severe hemorrhagic fever or shock. The illness begins with a febrile phase lasting up to 1 week, followed by a nonfebrile critical phase during which patients either improve or progress to severe dengue. Warning signs of progression to the severe disease state include persistent vomiting, abdominal pain, mucosal bleeding, respiratory distress, early signs of shock, and thrombocytopenia with an increase in the hematocrit level. Severe disease is manifested by increased vascular permeability with plasma leakage leading to pleural effusions, ascites, hypovolemic shock, and hemorrhage. Laboratory confirmation depends on the phase of illness and prior infection status.

Hospitalization is recommended for patients demonstrating warning signs or evidence of severe disease. When warning signs are not present, daily outpatient assessment is recommended.

Treatment is supportive, as no antiviral treatment exists for dengue. Supportive therapy includes hydration, avoidance of aspirin and nonsteroidal anti-inflammatory drugs (NSAIDs) to minimize the risk of bleeding, and frequent monitoring for warning signs of severe disease or dehydration. Management of severe disease includes maintaining hemodynamic stability until plasma leakage resolves (World Health Organization (WHO), Dengue: Guidelines for diagnosis, treatment, prevention and control. 2009).

5:

In the absence of warning signs of severe dengue, patient may be managed in the home setting with daily follow-up provided there is evidence of adequate oral intake and urine output. The following factors warrant observation in the hospital setting (World Health Organization (WHO), Dengue: Guidelines for diagnosis, treatment, prevention and control. 2009):

- Old age
- Inadequate oral intake
- Comorbidity
- Lack of access to emergency services
- Unreliable follow-up care

6:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. These rates should be increased in the presence of dehydration or insensible loss. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

7:

Unreliable follow-up refers to a low likelihood of the patient returning for outpatient reassessment after discharge. To be deemed unreliable, there must be a history of non-compliance with keeping appointments or a logistical barrier (e.g., lack of transportation, living remote from outpatient care).

8:

In a review of 7 randomized controlled trials on the use of corticosteroids for the treatment of patients with mononucleosis, a Cochrane review found one trial that reported a decrease in the inpatient length of stay, but did not specify its role in symptom management (Rezk et al., Cochrane Database Syst Rev 2015: CD004402).

9:

A peritonsillar abscess is an abscess which forms between the tonsil and the pharynx. It is usually unilateral and is often associated with tonsillitis. Symptoms may include fever, sore throat, trouble swallowing, and ear pain. Treatment with antibiotics and abscess drainage, either surgically or by needle aspiration, is usually successful. If untreated, complications include airway compromise and aspiration pneumonia. Patients with these complications may need to be treated at the Critical level of care (Galioto, Am Fam Physician 2017, 95: 501-6; Chang et al., Cochrane Database Syst Rev 2016, 12: CD006287).

10:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

11:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

12:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

13:

A central venous catheter is a catheter with a tip that lies within a central vein (superior vena cava, inferior vena cava, subclavian, brachiocephalic, internal jugular, iliac, or femoral veins). Catheters may include peripherally inserted central venous catheters (PICC), non-tunneled central venous catheters, tunneled central venous catheters, and totally implantable catheters (O'Grady et al., Clin Infect Dis 2011, 52: e162-93; Rupp et al., Anesthesiology 2012, 116: 539-73).

14:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

15:

In patients with central venous catheters, purulence is a sign of a complicated exit site infection requiring empirical systemic anti-infective therapy. In the absence of purulence or systemic signs of infection, topical anti-infective therapy is sufficient (Mermel et al., Clin Infect Dis 2009, 49: 1-45).

16:

Warning signs indicate a high risk of progression to severe dengue and warrant hospitalization (World Health Organization (WHO), Dengue: Guidelines for diagnosis, treatment, prevention and control. 2009). While some studies question the effectiveness and applicability of risk stratification according to the warning signs (Leo et al., BMC Infect Dis 2013, 13: 498; Thein et al., PLoS Negl Trop Dis 2013, 7: e2023; Barniol et al., BMC Infect Dis 2011, 11: 106; Hadinegoro, Paediatr Int Child Health 2012, 32 Suppl 1: 33-8), the evidence-base for dengue is evolving and further investigation is needed to confirm these findings (Horstick et al., Pathog Glob Health 2012, 106: 94-101).

17:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

18:

The Glasgow Coma Scale (GCS) is considered to be the old standard for documenting level of consciousness. The 3 areas of functioning that are examined are eye opening, best motor response and verbal response. Scoring ranges from a low of 3 to a high of 15. Scoring is indicative of the following level of consciousness; ≥ 8 represents severe brain injury and coma, a score between 9 and 12 indicates moderate brain injury and a score from 13 to 15 is indicative of mild brain injury.

19:

Petechiae are small, superficial deposits of extravasated blood in the sub-epidermal and dermal skin and range from 1 mm to 5 mm in diameter. Unlike the erythema of dilated capillaries in vascular lesions, petechiae characteristically do not blanch with direct pressure.

20:

Protracted vomiting refers to vomiting that is still active despite multiple doses of parenteral antiemetics.

21:

Fever of unknown origin (FUO) is classically defined as a documented temperature of 101.0°F (38.3°C) or greater on multiple occasions over a three week period without an identifiable etiology despite a thorough investigation. Hospitalization may be required to rule out a life-threatening cause (e.g., infection, malignancy) or in the setting of clinical instability (Hersch and Oh, Am Fam Physician 2014, 90: 91-6).

22:

The outpatient evaluation of prolonged fever is considered non-diagnostic when the etiology remains elusive despite a comprehensive diagnostic work-up including (Hersch and Oh, Am Fam Physician 2014, 90: 91-6):

- History and physical examination
- Complete blood count
- Urinalysis and urine culture
- Erythrocyte sedimentation rate and c-reactive protein
- Blood chemistry screen (e.g., electrolytes, liver enzymes)
- Immunologic testing (e.g., antinuclear antibodies, rheumatoid factor)
- Serologic testing (e.g., human immunodeficiency virus, cytomegalovirus)
- Tuberculosis evaluation (e.g., purified protein derivative, interferon-gamma release assay)
- Blood cultures
- Chest radiograph

23:

Infections of the eye may include endophthalmitis, severe bacterial corneal ulcers, and hyperacute bacterial conjunctivitis.

24:

Surgical biopsy refers to any specimen obtained by either invasive or surgical procedure involving internal organs (e.g., lungs, stomach), sinus, pleura, pericardium, lymph nodes, etc.

25:

Effluent is the liquid by-product of any internal organ eliminated through a surgically placed drain. It is generally applied only to liquids that are expected to be sterile at the point the drain is installed. It is not applied to liquids that are externally collected through normal body excretions (e.g., feces, clean-catch urine, etc.).

26:

Recommendations regarding evaluation, treatment, and setting of care for patients with fever and neutropenia are based on risk stratification into low and high risk groups. Assessment of the risk for complications from a severe infection should be performed upon presentation.

Low risk patients are generally defined by clinical stability without an active medical comorbidity (e.g., mucositis, nausea and vomiting, mental status change, etc.). High risk factors may include, but are not limited to, the following:

- Absolute neutrophil count (ANC) $\leq 100/\text{cu.mm}(100 \times 10^6/\text{L})$
- Mental status change
- Chronic lung disease
- Abdominal pain, new onset
- Mucositis
- Nausea and vomiting
- Diarrhea
- Uncontrolled pain
- Underlying malignancy with bone marrow involvement (e.g., acute leukemia)

In addition, use of validated risk assessment tools which take into account a multitude of risk factors, such as the Multinational Association for Supportive Care in Cancer (MASCC) scoring system, are recommended to categorize patients into low and high risk groups. Pediatric guidelines specifically recommend that hospitals adopt and use a validated risk stratification tool to guide routine management. Select low risk patients may be candidates for outpatient management, but inpatient management remains the standard of care (Pettit et al., Am J Emerg Med 2021, 50: 5-9; Zimmer and Freifeld, J Oncol Pract 2019, 15: 19-24).

27:

The absolute neutrophil count (ANC) is the percentage of available polymorphonuclear cells (also referred to as polys, PMNs, or segs) and bands multiplied by the white blood cells (WBC). The resulting count predicts the body's ability to respond to an infection. Neutropenia is the condition of a low ANC. Patients with an ANC less than 500/cu.

mm($500 \times 10^6/L$) are severely neutropenic and have a very limited immunologic response to infection. To determine the ANC, a differential WBC count must be performed. Segs and bands may be reported as a percentage of the WBC or can be reported in total number.

The formula for calculating an ANC is neutrophils (segs + bands) $\times$ WBC.

Examples:

ANC calculation using a percentage of total WBC count:

WBC count: 1,500/cu.mm($1.5 \times 10^9/L$)

Segs: 20%(0.20) of the WBCs

Bands: 2%(0.02) of the WBCs

$(20\% + 2\%(0.20 + 0.02)) \times 1,500/\text{cu.mm} (1.5 \times 10^9/L)$

$= 330/\text{cu.mm} (330 \times 10^6/L)$

Normal range: 1,500-8,000/cu.mm($1.5-8.0 \times 10^9/L$)

An ANC of less than 500/cu.mm($500 \times 10^6/L$) is considered severely neutropenic.

ANC Calculation using the above numbers as whole numbers:

WBC count: 1.5

Segs: 0.2

Bands: 0.02

$(0.2 + 0.02) \times 1.5 = 0.33$

Normal range: 1.5 to 8.0

An ANC of less than 0.5 is considered severely neutropenic.

28:

An absolute neutrophil count that is expected decline to $< 500/\text{cu.mm} (500 \times 10^6/L)$ within 48 hours is included in the definition of neutropenia (National Comprehensive Cancer Network, The NCCN clinical practice guidelines in oncology, Prevention and Treatment of Cancer-Related Infections (Version 2.2023). 2023).

29:

Patients with a retropharyngeal abscess and a risk of airway compromise may require admission to the Critical level of care.

30:

Public health risk means that the patient has demonstrated an inability or unwillingness to follow through with a medication regimen and respiratory precautions necessary to prevent the spread of disease to significant others and the community at large.

31:

Care is recommended in a negative pressure room for patients with suspected drug-resistant tuberculosis (TB), until nonresistance is confirmed. In confirmed multidrug-resistant TB, a negative pressure room should be utilized until there are 3 negative acid-fast bacillus smears at weekly intervals and ideally a negative culture. For patients with confirmed multidrug-resistant TB, earlier discharge may be considered when there are adequate facilities for home isolation and the patient will adhere to treatment. Arrangements need to be made for supervising and administering all anti TB therapy (NICE, 2016 (revised 2019 Sept), Clinical guideline; no. [NG33]).

32:

Typhoid fever is an infectious disease caused by *Salmonella enterica* serotype *Salmonella typhi*. The disease is transmitted through ingestion of contaminated food or water. The severity of illness ranges from mild to severe, and may be divided into non-complicated and complicated cases. Non-complicated disease is manifested by prolonged fever, bowel dysfunction (e.g., constipation in adults and diarrhea in children), headache, general malaise, and loss of appetite. A dry cough may be present as well. Complicated, or severe, disease is characterized by increasing abdominal pain along with gastrointestinal bleeding or intestinal perforation. A diagnosis of typhoid fever is confirmed by a positive culture. In some cases, a definitive diagnosis cannot be made and suspicion relies on a combination of clinical findings and the detection of a specific antibody response. Treatment consists of anti-infective therapy along with supportive measures (e.g., hydration, nutrition). Most patients are managed as outpatients with oral anti-infective therapy and close follow-up. Hospitalization for parenteral anti-infective therapy may be warranted for patients with persistent vomiting, severe diarrhea, and abdominal distension. In addition, mental status changes require close evaluation and high-dose corticosteroid therapy for suspected

meningitis. Choice of anti-infective agent is dependent on local susceptibility due to variable resistance patterns (World Health Organization (WHO), Background document: The diagnosis, treatment and prevention of typhoid fever. 2003; Bhan et al., Lancet 2005, 366: 749-62).

33:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

34:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

35:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2 . The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

36:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

37:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination

of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

38:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Semler and Rice, Clin Chest Med 2016, 37: 241-50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).

Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

39:

Patients with sepsis-induced tissue hypoperfusion can present with elevated blood lactate levels greater than 2 mmol/L (18.0 mg/dL) (Shankar-Hari et al., JAMA 2016, 315: 775-87).

40:

Devices utilized for invasive hemodynamic monitoring include an arterial line, a central venous catheter, or a pulmonary artery catheter (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408; McKinley et al., Shock 2009, 32: 463-70; Wheeler et al., N Engl J Med 2006, 354: 2213-24).

41:

Instruction: If infection is listed below, choose the appropriate Infection subset. If there is no subset for the specific infection suspected or diagnosed, continue the review using the current subset.

42:

Instruction: Selection of this criteria is appropriate for patients who have a symptom or finding not associated with a condition specific subset. Refer to the review process or subset level note for more information.

43:

Management of dengue in the absence of warning signs includes oral or intravenous fluid intake and close monitoring of vital signs, fluid balance, and hematocrit levels (World Health Organization (WHO), Dengue: Guidelines for diagnosis, treatment, prevention and control. 2009).

44:

The recommended frequency of hematocrit monitoring is at least every 6 to 12 hours (World Health Organization (WHO), Dengue: Guidelines for diagnosis, treatment, prevention and control. 2009).

45:

Isotonic fluid administered intravenously (IV) is indicated in patients with dengue when warning signs are present. IV fluid should be titrated according to perfusion status and urine output, beginning at a rate of 5-7 mL/kg/h with a gradual reduction to 2-3 mL/kg/h. Fluid rates are based on ideal body weight (World Health Organization (WHO), Dengue: Guidelines for diagnosis, treatment, prevention and control. 2009).

46:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

47:

Peritonsillar abscesses may require additional time to mature before surgical drainage is appropriate.

48:

Glasgow Coma Scale (GCS) is a tool commonly used to measure neurological function. GCS evaluates the level of consciousness in acutely ill patients. A patient's GCS is the combined total score accumulated through the evaluation of the following 3 areas (ranging from 3-15) (Chou et al., In: Glasgow Coma Scale for Field Triage of Trauma: A Systematic Review. 2017):

- Eye opening - 4 = Spontaneous, 3 = To voice, 2 = To pain, 1 = None
- Verbal response - 5 = Oriented, 4 = Confused, 3 = Inappropriate, 2 = Incomprehensible sounds, 1 = None
- Motor response - 6 = Obeys verbal commands, 5 = Localizes to pain, 4 = Withdraws to pain, 3 = Flexion response to pain, 2 = Extension response to pain, 1 = None

Of note, age may impact the relationship between anatomic injury severity and GCS scores such that the same GCS score may correspond to a greater severity of traumatic brain injury (TBI) in adults aged 65 and older than in those younger than 65 (Salottolo et al., JAMA Surg 2014, 149: 727-34). For patients less than 5 years of age, the verbal response component includes a modified scoring as follows:

- 5 = Appropriate word or social smile / Coos and babbles, 4 = Cries but consolable, 3 = Cries, screams, persistently irritable, 2 = Restless, agitated, moans, 1 = None

49:

Management of a patient with warning signs includes frequent monitoring of vital signs, peripheral perfusion, fluid balance, and hematocrit levels. Administration of isotonic fluid is recommended to maintain adequate perfusion and urine output. Additional laboratory testing is indicated based on history and physical exam (World Health Organization (WHO), Dengue: Guidelines for diagnosis, treatment, prevention and control. 2009).

50:

The secondary evaluation should be guided by suspicions raised during the initial work-up and may include repeated testing to investigate trends, as well as more advanced testing such as specialized laboratory, imaging, and invasive diagnostic studies. Examples of advanced testing include computed tomography scans, magnetic resonance imaging, and biopsies (Hersch and Oh, Am Fam Physician 2014, 90: 91-6).

51:

A fever is suspected as being new onset when a patient has been afebrile and then develops a new fever or when the fever pattern changes.

52:

Fevers that require additional work up or treatment in the hospitalized patient that cannot be followed up appropriately as an outpatient, may be reviewed in this subset. If a condition specific subset exists that covers this fever (e.g., Pneumonia, DVT) the patient should be reviewed in the appropriate condition specific subset.

53:

A drug-induced fever is a non-infectious febrile response following the administration of a drug that resolves when the drug is discontinued. The patient is often well-appearing and demonstrates resolution of the underlying disease process, despite the presence of a fever. A drug-induced fever should be suspected when the cause of the fever remains unknown, despite a full diagnostic work-up, or when the patient is receiving a drug frequently associated with fever. The most common agents triggering fever include anticonvulsants, anti-infectives, cardiac medications, allopurinol, and heparin.

The following drugs are also capable of inducing hyperthermia (Walter and Carraretto, Journal of the Intensive Care Society 2015):

- Serotonin-noradrenalin reuptake inhibitors
- Selective serotonin reuptake inhibitors
- Monoamine oxidase inhibitors
- Tricyclic antidepressants
- Bupropion

- Opioids
- Central nervous system stimulants
- Anticholinergics
- Antipsychotics
- Antispasmodics
- Sympathomimetic agents

Treatment involves discontinuing medications that are unnecessary, recently initiated, or highly associated with inducing fever. Resolution of the fever should occur within 48 to 72 hours of discontinuing the precipitating agent (Patel and Gallagher, Pharmacotherapy 2010, 30: 57-69; Eyer and Zilker, Crit Care 2007, 11: 236).

54:

A neurological assessment establishes a baseline so that subtle changes can be monitored. A comprehensive neurological assessment often includes an evaluation of:

- Mental status (e.g., level of consciousness, orientation, insight, calculation ability)
- Cranial nerves (e.g., olfactory, optic, vestibulocochlear)
- Motor system (e.g., muscle tone, strength, reflexes)
- Sensory system (e.g., light touch, pain, temperature)
- Coordination (e.g., orchestration and fluidity of movement)
- Gait (e.g., heel-to-toe straight-line walking)

55:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

56:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

57:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

58:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

59:

Oral intake should be greater than or equal to the patient's maintenance fluid requirements or within parameters which the medical practitioner determines are acceptable. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid requirements (*mL/kg/d*) can be calculated based on current body weight (*kg*):

- 100 *mL/kg/d* for first 10 *kg*
- 50 *mL/kg/d* for second 10 *kg*

- 20 mL/kg/d for each additional kg in weight

60:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit.

Maintenance fluid rates can be determined using the 4-2-1 calculation formula, which is based on ideal body weight in kilograms. To convert pounds to kilograms:

- weight (lbs) divided by 2.2 = weight (kg)

To calculate the fluid rate:

- For the first 10 kg at 4 mL/kg/h = 40 mL/h
- For the second 10 kg at 2 mL/kg/h = 20 mL/h
- For each kg over 20 kg = 1 mL/h up to 125mL/h

61:

In most patients hospitalized with infection, clinical improvement would be expected to occur within 2 days. For patients who are deteriorating or not responding to treatment, it may be necessary to broaden anti-infective coverage while awaiting the results of additional diagnostic tests. In addition, consultation by an infectious disease specialist and further diagnostic tests may be valuable to evaluate reasons for the lack of improvement.

62:

Resolution of infection is indicated by a decrease in temperature, a normalizing of white blood cells and other lab values, and improving imaging studies. If there is a wound present, signs of infection should be resolving (e.g., decrease in wound drainage, less pain at site).

63:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360:

j5790).

64:

Patients with pulmonary tuberculosis who have been deemed medically stable, including those with a positive acid-fast bacillus (AFB) sputum smear, are considered safe for discharge if the following parameters have been met:

- Follow-up care with a local tuberculosis-control program has been arranged
- A multi-drug tuberculosis regimen has been initiated
- They do not share a household with an immunocompromised person, infant, or child less than four years old
- All immunocompetent people that reside in the household have been previously exposed to the patient
- The patient is willing to not travel outside the home (except for healthcare visits) until they have obtained a negative acid-fast bacillus (AFB) sputum smear

65:

Relevant complications or comorbidities are conditions that actively require management during an inpatient stay.

66:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirement. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit.

67:

Anti-infective medications may cause adverse drug reactions. Symptoms of an adverse drug reaction include urticaria, angioedema, bronchospasm, pruritus, rash, vomiting, diarrhea, anaphylaxis, and thrombocytopenia.

68:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

69:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

70:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

71:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

72:

Signs indicating recovery from *Clostridioides difficile* include:

- Decrease in the severity and frequency of diarrhea
- Absence of fever
- Normal white blood cell count

Not all patients demonstrate fever or leukocytosis during the course of illness. Diet advancement should be based on improving clinical signs and symptoms.

73:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

74:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives. Guidelines describe multiple acceptable regimens but do indicate a preference for oral fidaxomicin for patients with mild to moderate disease. Alternative regimens are driven by disease severity and the number of episodes the patient has experienced. Treatment guidelines recommend discontinuation of the inciting anti-infective as soon as possible to prevent recurrence. Other interventions include IV fluids, electrolytes, or oral fluids to maintain hydration. Use of opiates and antidiarrheal medications should be avoided because decreased intestinal motility will exacerbate the infection (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

75:

Instruction: This criteria point is intended for anti-infectives initiated for the treatment of *C. difficile* infections. The preferred treatment of *Clostridioides difficile* infections is based on the episode of illness and the severity of the symptoms. The Infectious Diseases Society of America (IDSA) recommends a variety of treatment regimens which are dependent on clinical presentation, including disease severity and how many episodes of disease the patient has experienced (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

76:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit (Scales, Br J Nurs 2014, 23).

77:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

78:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

79:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g.,

secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

80:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

81:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

82:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

83:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home

would be cumbersome and places the patient at risk

- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

84:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

85:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

86:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

87:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the

need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

88:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

89:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

90:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

91:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

92:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

93:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

94:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

95:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

96:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

97:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive

- Inspiratory muscle weakness

98:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.